Stem Cells: Evidence Before Judgement

LAUNCH · Science · 40 minutes

I can describe stem-cell functions and distinguish scientific evidence, ethical concern and uncertainty.

Prepare before pupils arrive

Setup: Prepare the three-column mat, three neutral statement cards and a visible glossary. Check every medical claim is cautious and generic.

Safety: Do not frame disability as tragedy or cure as the only valuable outcome. Keep discussion optional and impersonal.

Equipment: Stem-cell type cards, evidence/ethics/uncertainty mat, glossary sheet and pencil.

Safety: Use neutral, non-personal cases. No pupil is asked to disclose beliefs, fertility history, disability or medical experience; pass is always available.

Fallback: Point to or direct an adult to place each card, then dictate the scientific reason. Exact adult scribing is equivalent evidence.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Discuss potential benefits and risks associated with the use of stem cells in medicine.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 4 min	State one role of mitosis in growth or repair. Describe differentiation in one sentence.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today’s answer.
Starter · 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask ‘What makes you think that?’ and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 5 min	stem cell — unspecialised starting cell differentiate — become specialised ethical — about what ought to be done Begin with function: stem cells can divide, then some daughter cells can differentiate into specialised cells. Distinguish sources: embryonic stem cells can form many cell types; adult stem cells usually form a more limited range; meristems produce new plant cells.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 5 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Scientific evidence can estimate possible benefit and harm; it cannot decide a moral value by itself. An ethical concern may focus on the source of cells, consent, fairness or acceptable harm.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Place each statement by its main role. Then explain why one statement could matter to more than one column.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 10 min	Build a stem-cell evidence glossary and classify three short statements before writing one cautious scientific claim. Record: An accurate glossary, classified statements and one cautious claim that distinguishes possibility from evidence.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Distinguish scientific evidence from an ethical concern in this context. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Give one next move; the pupil edits or re-attempts on the existing work.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: It can divide and produce cells that differentiate, supporting growth or replacement.

Yes. The statement says what the cell can do and links that function to a biological outcome.

Misconception repair

Repaired. Ability to differentiate explains a possible mechanism, not proof that a treatment is safe or effective.

Guided checkpoint

Yes. The statement says what the cell can do and links that function to a biological outcome. A function states an action or role, not only a label and not an unsupported treatment promise.

Observed division

SCIENTIFIC EVIDENCE. Evidence: this is an observation from a defined investigation. Ask whether the statement reports an observation, a value or a gap in knowledge.

Consent matters

ETHICAL CONCERN. Ethical concern: this asks what ought to be permitted and how people should be treated. Look for a question about rights, values or responsibilities.

Long-term outcome unknown

UNCERTAINTY. Uncertainty: the evidence cannot yet answer the long-term question. Look for information that the investigation has not yet provided.

Model limitation

Real decisions involve more evidence, different stakeholders and changing medical knowledge; three cards cannot settle an ethical question.

Independent evidence

An accurate glossary, classified statements and one cautious claim that distinguishes possibility from evidence.

Supported: Match stem cell, differentiate and ethical to icon-backed meanings; sort three statements. Keep the word bank visible and use: This is __ because it says __.

Standard: Classify three statements and describe one stem-cell function before writing one cautious claim. Use: Evidence shows __. It does not yet show __.

Stretch: Classify three statements, identify one overlap and explain why function alone cannot prove treatment success. Name evidence, value and uncertainty separately; avoid adding Higher-tier content.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_L_W11L1_Stem_Cell_Evidence_Introduce.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.